

Discussion Assignment 8

Dr. Parallelepiped or: How I learned to Stop Worrying and Love Pronouncing Long Words

MATH 2210, Spring 2018

Background

As we saw with double integrals, we may be able to simplify a triple integral of the form

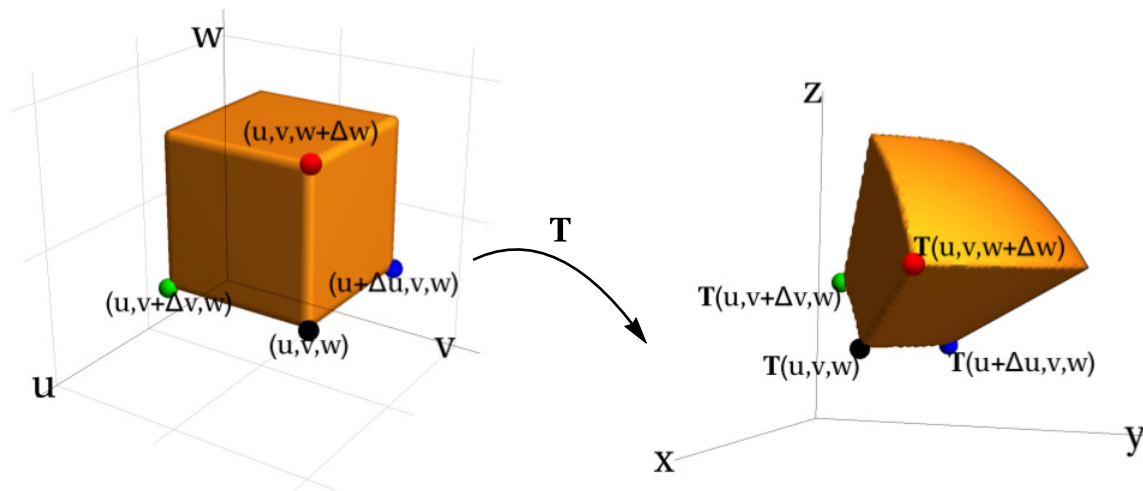
$$\iiint_{\mathcal{D}} f(x, y, z) \, dV$$

by making a change of variables (that is, a substitution) of the form

$$\mathbf{T} : x = g(u, v, w), \quad y = h(u, v, w) \quad \text{and} \quad z = p(u, v, w).$$

where x, y , and z are functions of new variables u, v , and w . This transformation maps a region $\mathcal{R}$ in the uvw -space to a region $\mathcal{D}$ in the xyz -space. The equations $x = g(u, v, w)$, $y = h(u, v, w)$ and $z = p(u, v, w)$ convert u, v , and w to x, y , and z ; we call these formulas the change of variable formulas.

To complete the change to the new u, v, w variables we need to understand the volume element, dV , in this new system. To accomplish this we partition the uvw -space into small boxes with width Δu , depth Δv , and height Δw . When we transform one of these boxes into the standard xyz -space we get a new “curvy box.”



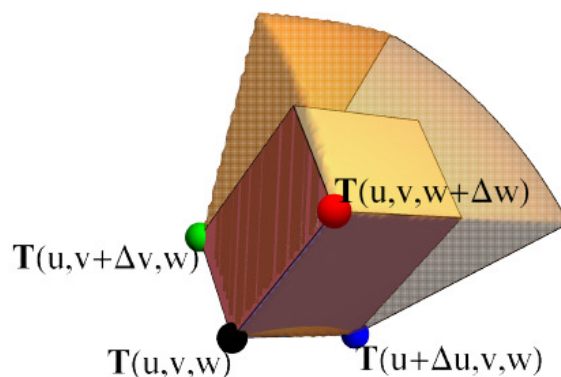
The volume of the “curvy boxes” in the xyz -space is important to the triple integral because it is the base of the 4 dimensional box we use to approximate the 4 dimensional volume calculated by a triple integral. Assuming that the transformation $\mathbf{T}$ is differentiable and the sizes Δu , Δv , and Δw are sufficiently small then the “curvy region” can be thought as a parallelepiped.

The vectors that span the parallelepiped are

$$\mathbf{T}(u + \Delta u, v, w) - \mathbf{T}(u, v, w)$$

$$\mathbf{T}(u, v + \Delta v, w) - \mathbf{T}(u, v, w)$$

$$\mathbf{T}(u, v, w + \Delta w) - \mathbf{T}(u, v, w)$$



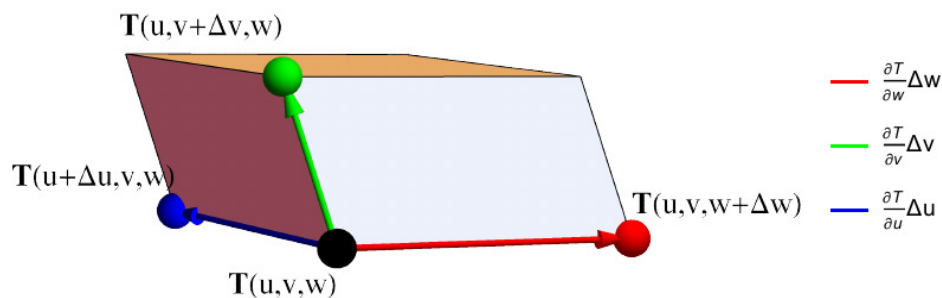
We can approximate these vectors as:

$$\mathbf{T}(u + \Delta u, v, w) - \mathbf{T}(u, v, w) = \frac{\mathbf{T}(u + \Delta u, v, w) - \mathbf{T}(u, v, w)}{\Delta u} \Delta u \approx \frac{\partial \mathbf{T}}{\partial u}(u, v, w) \Delta u$$

$$\mathbf{T}(u, v + \Delta v, w) - \mathbf{T}(u, v, w) = \frac{\mathbf{T}(u, v + \Delta v, w) - \mathbf{T}(u, v, w)}{\Delta v} \Delta v \approx \frac{\partial \mathbf{T}}{\partial v}(u, v, w) \Delta v$$

$$\mathbf{T}(u, v, w + \Delta w) - \mathbf{T}(u, v, w) = \frac{\mathbf{T}(u, v, w + \Delta w) - \mathbf{T}(u, v, w)}{\Delta w} \Delta w \approx \frac{\partial \mathbf{T}}{\partial w}(u, v, w) \Delta w$$

When we label the edges of the parallelepiped and turn it (just to show it at another angle) we get:



Therefore the volume of the “curvy boxes” is approximately equal to the area of the parallelepiped and we get the following statement:

$$\Delta V \approx \text{parallelepiped the volume}$$

Question

Find the volume of the parallelepiped spanned by the vectors

$$\frac{\partial \mathbf{T}}{\partial u}(u, v, w)\Delta u, \quad \frac{\partial \mathbf{T}}{\partial v}(u, v, w)\Delta v, \quad \text{and} \quad \frac{\partial \mathbf{T}}{\partial w}(u, v, w)\Delta w$$

and choose how the triple integral in the new variables u, v , and w compares to the integral in the old variables x, y , and z .

$$\iiint_{\mathcal{D}} f(x, y, z) \, dx dy dz =$$

$$(a) = \iiint_{\mathcal{R}} f(u, v, w) \left| \frac{\partial x}{\partial u} \left(\frac{\partial y}{\partial v} \frac{\partial z}{\partial w} - \frac{\partial y}{\partial w} \frac{\partial z}{\partial v} \right) - \frac{\partial x}{\partial v} \left(\frac{\partial y}{\partial u} \frac{\partial z}{\partial w} - \frac{\partial y}{\partial w} \frac{\partial z}{\partial u} \right) + \frac{\partial x}{\partial w} \left(\frac{\partial y}{\partial u} \frac{\partial z}{\partial v} - \frac{\partial y}{\partial v} \frac{\partial z}{\partial u} \right) \right| \, du dv dw$$

$$(b) = \iiint_{\mathcal{R}} f(u, v, w) \left| \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial y}{\partial u} \frac{\partial x}{\partial v} \right| \, du dv dw$$

$$(c) = \iiint_{\mathcal{R}} f(u, v, w) \left| \left\langle \frac{\partial y}{\partial u} \frac{\partial z}{\partial v} - \frac{\partial z}{\partial u} \frac{\partial y}{\partial v}, - \left(\frac{\partial x}{\partial u} \frac{\partial z}{\partial v} - \frac{\partial z}{\partial u} \frac{\partial x}{\partial v} \right), \frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial y}{\partial u} \frac{\partial x}{\partial v} \right\rangle \right| \, du dv dw$$

$$(d) = \iiint_{\mathcal{R}} f(u, v, w) \left| \left\langle \frac{\partial x}{\partial u} \left(\frac{\partial y}{\partial v} \frac{\partial z}{\partial w} - \frac{\partial y}{\partial w} \frac{\partial z}{\partial v} \right), - \frac{\partial x}{\partial v} \left(\frac{\partial y}{\partial u} \frac{\partial z}{\partial w} - \frac{\partial y}{\partial w} \frac{\partial z}{\partial u} \right), \frac{\partial x}{\partial w} \left(\frac{\partial y}{\partial v} \frac{\partial z}{\partial u} - \frac{\partial y}{\partial u} \frac{\partial z}{\partial v} \right) \right\rangle \right| \, du dv dw$$

Answer 1: _____

Answer 2: _____

Answer 3: _____

Individual Work

Now you are to write a report, three page maximum, that convinces an audience of your peers that you understand how to represent the volume of the parallelepiped. This report will need to be saved as a pdf and submitted to your discussion sections WyoCourses page. A scoring rubric and guideline on writing good mathematics is posted on our WyoCourses page.